

AI Work Portal

A NAS-native workspace for multimodal AI

Unify file intake, meeting recording, speech transcription, transcript translation, document OCR, vector retrieval, generative models, and complete auditability from one operational portal.

1

NAS data sovereignty

Keep source assets, indexes, vectors, and audits together

2

Local and cloud AI

Choose execution location and model per workload

3

Traceable automation

See discovery, processing state, source, and model end to end

All persistent information is stored on the NAS

Source files, transcripts, translations, document chunks, vectors, answer cache, model weights, and call records remain on NAS storage. Content leaves the NAS only when a user explicitly selects a cloud model for that operation.

CORE CAPABILITIES

24×7

NAS intake watch

8

Core AI workflows

2-tier

Answer cache

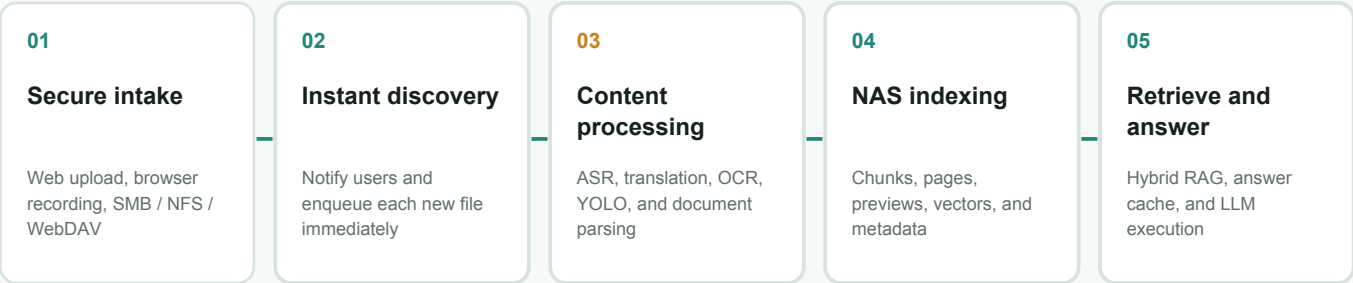
ZH / EN

Bilingual UI

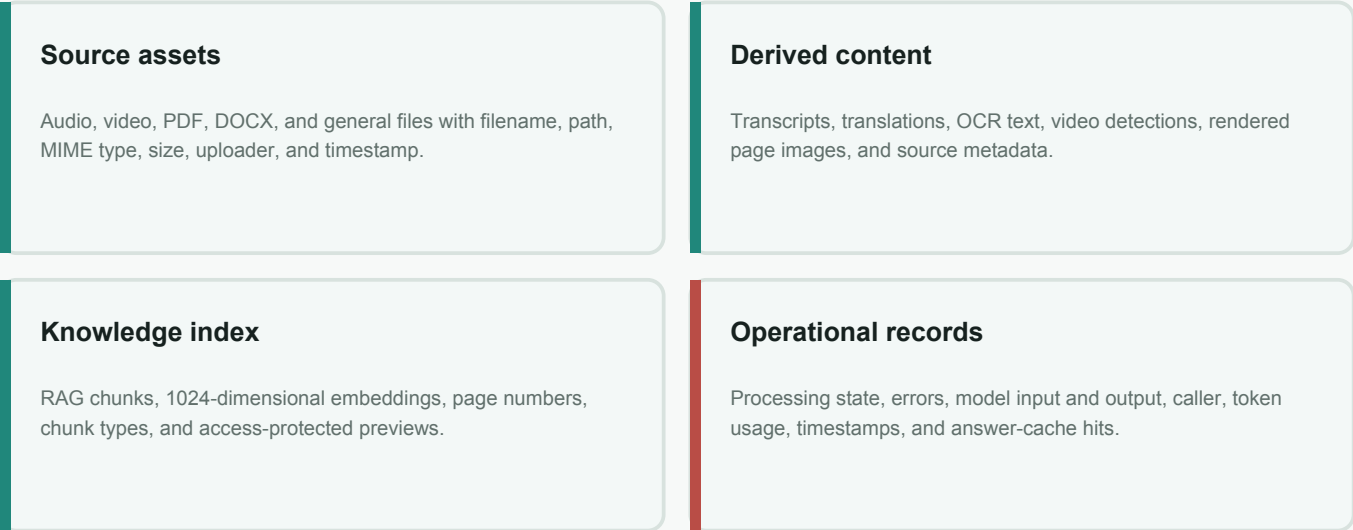
A NAS-first data lifecycle

DATA SOVEREIGNTY / GOVERNANCE / LIFECYCLE

AI Work Portal brings shared storage, access control, AI processing, and knowledge search into one entry point. Every file is assigned a NAS asset ID that links its source, processing stages, model output, and audit trail.



What stays on the NAS



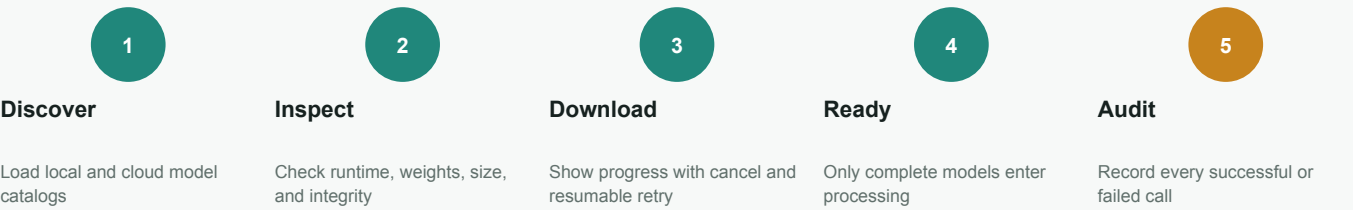
Access, versioning, and live status

- Login and session cookies protect asset APIs, recordings, and source-page images.
- Users see their own data; administrators manage the full asset scope with a clear integration point for NAS ACLs.
- AI output never overwrites source files, supporting reprocessing, version comparison, and NAS snapshots.
- WebSocket updates report intake, processing, completion, missing-model, and failure states in real time.

Automatic model discovery, download, and management

MODEL OPERATIONS / LOCAL AI / CLOUD CHOICE

The portal automatically reads the model catalog and local weight state, detecting runtime readiness, downloaded size, integrity, and availability. Downloads remain user-initiated so NAS capacity and network usage stay controlled.



MODEL PORTFOLIO

Speech / ASR	whisper.cpp small / base, faster-whisper, SenseVoiceSmall, and Paraformer-zh; cloud options include OpenAI Transcribe, Deepgram, and AssemblyAI.
Translation and generation	Local Qwen3-4B-Q4_K_M on the NAS; cloud choices include OpenAI, Claude, Gemini, DeepSeek, Qwen, Mistral, Groq, and more.
Embeddings	Qwen3-Embedding-0.6B runs on CPU and creates 1024-dimensional vectors for documents and queries.
Vision and OCR	YOLO frame sampling and object detection; PyMuPDF page rendering with PaddleOCR for image-based PDFs.

NAS Model Management panel			
Installation state	Capacity details	Operator controls	Runtime isolation
Installed, partial, missing, downloading, cancelled, or failed	Expected size, downloaded bytes, model path, and progress percentage	Download, cancel, retry, and refresh with partial-file resume support	Separate dependency and device management for web, generation, embedding, ASR, and vision

Multimodal processing, Hybrid RAG, and specifications

MULTIMODAL / HYBRID RAG / AUDIT

Meetings and audio

Preserve source voice; choose local or cloud ASR; then choose a target language and translation LLM; index source and translated text separately.

PDF / DOCX

Use pypdf for text PDFs; render image PDFs with PyMuPDF and read them with PaddleOCR; show page citations and previews with answers.

Video

Sample frames with YOLO and store objects, confidence, timestamps, and previews as video_detection chunks.

AI Work

Summarize, query, and analyze NAS assets with local Qwen or mainstream cloud models, including pricing and API-key requirements.

Hybrid RAG and cost control

75%	25%	0.76	2-tier
Vector similarity	Keyword score	Semantic cache threshold	Answer cache
Qwen3 Embedding + cosine similarity	Multilingual term matching	Reuse a relevant answer already held on the NAS	Normalized exact match + semantic match

Reference deployment

Web / API	FastAPI, HTTP 80 redirected to HTTPS 443, session login, and role-based access
Persistence	NAS filesystem + SQLite for assets, meetings, chunks, vectors, cache, and call audits
Local generation	Qwen3-4B-Q4_K_M / llama.cpp, OpenAI-compatible API, 4096-token context
Embedding	Qwen3-Embedding-0.6B, CPU, 1024 dimensions, independent service
Service control	systemd-managed portal and model services with startup and restart policies
Observability	Processing timeline, model state, errors, tokens, balance, and complete input/output records